

AURELIA • MAISON DE JOAILLERIE

High-End Luxury E-Commerce & Fine Gifting Mobile/Web Platform

CAPSTONE PROJECT SUMMARY REPORT

GROUP 13

Project Team & Contribution Matrix

Team Member	Primary Roles	Key Responsibilities & Deliverables
Sourabh Yadav	Backend & Architecture Engineering	Node.js/Express REST API design, MongoDB schema modeling, Firebase Auth integration, system data flow & API routing.
Sujal Warke	Backend and Frontend	Server endpoints implementation (/api/products, /api/orders, /api/cart), database seeding, Mongoose models, and error handling.
Pranay Kadam	Research and Data Gathering	Flutter client state architecture (Provider), responsive desktop/mobile layout routing, AuthGate workflow, and order tracking.
Sneha Hande	Frontend & UI/UX Design	Luxury design system (LuxuryTheme), bespoke gifting UI, animations (SplashScreen & Hero), Google Fonts typography, and color palette.

Summary of Contributions:

- Backend Development: **Sujal Warke & Sourabh Yadav**
- Frontend Development: **Sujal Warke & Sneha Hande**
- UI/UX Design & Theming: **Sneha Hande**
- Architecture Engineering & System Flow: **Sourabh Yadav & Sujal Warke**
- Research and Data Gathering: **Pranay Kadam**

1. Executive Summary

Aurelia • Maison de Joaillerie is a cross-platform luxury jewelry e-commerce and bespoke gifting application developed using Flutter 3.x, Dart, Node.js/Express, and MongoDB, with Firebase Authentication for identity management.

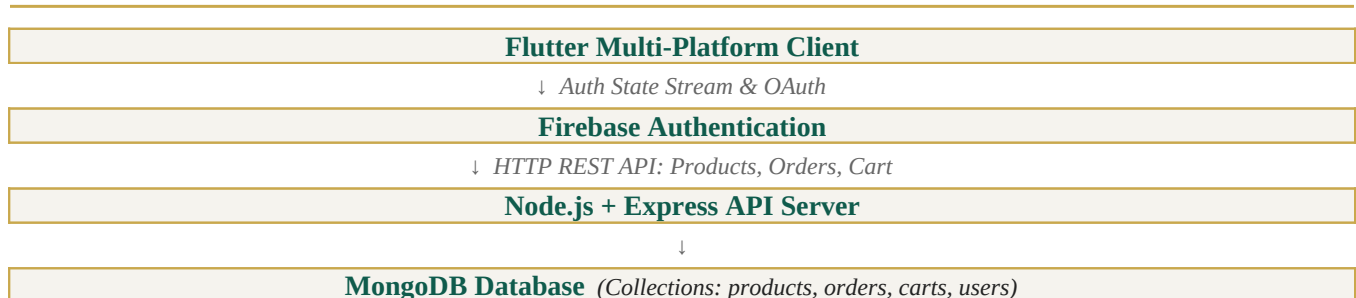
Unlike standard generic e-commerce applications, Aurelia is designed around the refined aesthetic of royal European jewelry houses. It addresses high-ticket fine jewelry retailing by integrating bespoke gifting options (handcrafted cedarwood boxes, royal velvet vaults, wax-seal calligraphy cards, and gold foil wrapping), simulated bank-grade escrow checkout verification, and an Executive Admin Dashboard for inventory curation and multi-stage order dispatch.

2. Technology Stack

- Frontend Client:** Flutter 3.x (compiled for Web, Android, iOS, and Desktop).
- Language:** Dart 3.x (Sound Null Safety).
- State Management:** Provider pattern (MultiProvider, ChangeNotifier).
- Authentication:** Firebase Auth (Email/Password & Google OAuth popup).

- **Backend Framework:** Node.js with Express.js (REST API on port 4000).
- **Database & ODM:** MongoDB with Mongoose.
- **Typography & Aesthetics:** Google Fonts (Cinzel, Playfair Display, Cormorant Garamond, Inter) paired with an Imperial Emerald (#0F5B4C) and Royal Gold (#C9A84C) palette.

3. System Architecture & Flow



1. Launch & Authentication Gate:

- *main.dart* initializes Firebase and provides three core state providers: *ShopProvider*, *AuthProvider*, and *ThemeProvider*.
- *SplashScreen* displays a curved scale-and-fade golden seal animation, then transitions to *AuthGate*.
- *AuthGate* intercepts session state: automatically routing administrators (*admin@aurelia.com*) to the Executive Lounge, returning consumers to their personalized boutique, and allowing seamless guest browsing.

2. Adaptive Client-Side Boutique:

- *MainRootScreen* detects screen dimensions via *MediaQuery*. Widths > 850px dynamically render an executive Left Sidebar Navigation (Desktop), while smaller screens render an ergonomic Bottom Navigation Bar (Mobile).
- *HomeScreen* incorporates slivers, category filter chips ("Rings", "Necklaces", "Earrings", "Bracelets"), price sliders, search filtering, and an adaptive 2, 3, or 4-column product grid with shared-element Hero transitions.

3. Bespoke Gifting & Cart Synchronizer:

- In *ProductDetailScreen*, consumers inspect gemstone/metal karat specifications and configure bespoke gifting add-ons.
- *CartScreen* calculates live breakdowns of base price, custom box upgrades, wax-seal calligraphy, and gold foil wrapping.
- Guest carts automatically synchronize with the server upon user login.

4. Escrow Checkout & Order Dispatch Tracking:

- *CheckoutScreen* enforces field validation via *GlobalKey<FormState>()* and triggers a 5-stage Bank Vault Escrow Handshake before generating a unique Vault Order Code (ORD-XXXXX).
- *OrderHistoryScreen* renders a real-time status tracker (Ordered → Shipped → Out for Delivery → Delivered).

5. Executive Admin Lounge:

- Tabbed administrative suite providing full CRUD capabilities (Add, Edit price/stock, Delete product) and live status updating for individual order items.

4. Module-Wise Work Breakdown

Backend Development (Sujal Warke & Sourabh Yadav)

- Designed and built the Express HTTP REST API across modular routes (/api/auth, /api/products, /api/orders, /api/cart).
- Structured Mongoose schemas with validation for Product, Order, Cart, and User entities.
- Implemented stock management logic ensuring inventory decrements accurately upon order placement.
- Seeded rich catalog data (seedProducts.js) featuring natural Zambian emeralds, brilliant diamonds, and 18K/22K gold jewelry.
- Constructed error-handling middlewares and CORS policies for web and mobile clients.

Frontend Development (Sujal Warke & Sneha Hande)

- Engineered client-side state handling via ShopProvider and AuthProvider.
- Built screens: HomeScreen, ProductDetailScreen, CartScreen, CheckoutScreen, OrderHistoryScreen, LoginScreen, and RegisterScreen.
- Implemented form validation logic, TextEditingController lifecycles, and TextFormField widgets.
- Connected HTTP services via ApiService to seamlessly fetch and mutate backend data.
- Programmed the comprehensive AdminDashboardScreen with inventory management tables and status modifiers.

UI/UX Design & Brand Styling (Sneha Hande)

- Established the Aurelia Sovereign Design System in LuxuryTheme (Emerald Green, Sovereign Forest Green, Royal Gold, Warm Sand, and Cream).
- Curated luxury typography pairings using GoogleFonts (Cinzel, Playfair Display, Cormorant Garamond, Inter).
- Designed custom reusable UI widgets: LuxuryDivider (gold filigree diamond separator), LuxuryButton (gold foil button), and WaxSealMessage (parchment letter).
- Designed light and dark mode canvas transitions managed by ThemeProvider.
- Implemented micro-interactions: explicit SplashScreen animations, curved transitions, and Hero animations.

Architecture Engineering & System Integration (Sourabh Yadav & Sujal Warke)

- Architected the Clean Layered Structure separating Presentation (screens/, widgets/), State (providers/), Domain (models/), and Services (services/).
- Designed the Hybrid Security Model: leveraging Firebase for authentication and identity while directing transactional e-commerce data through the Node.js/MongoDB pipeline.
- Engineered the Adaptive Responsive Layout Engine enabling seamless execution across mobile viewports and wide desktop displays.
- Implemented fail-safe offline resilience via DataService fallback data and socket error interception (_wrapNetworkErrors).

Research and Data Gathering (Pranay Kadam)

- Researched high-end jewelry retail conventions and bespoke gifting practices to inform product and feature scope.
- Gathered reference material for gemstone/metal specifications and catalog data used in seeding the product database.

- Compiled competitive and design benchmarking to guide the luxury aesthetic and UX decisions.

5. Key Project Accomplishments

-  **Zero-Crash Architecture:** Sound null safety throughout all Dart models and services.
-  **Cross-Platform Readiness:** Single codebase fully functional on Web, Desktop, and Mobile.
-  **Dual Persona Support:** Seamless consumer gifting workflow alongside an administrative executive portal.
-  **Luxury Brand Aesthetics:** Meets high-end visual design standards with smooth 60fps animations.

Report submitted for the Aurelia • Maison de Joaillerie Capstone Project.